SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Unit-wise MCQ Bank (with Answer Key)

Unit-1: Introduction to Networks

Course Outcome Mapped: CO-1 | Total MCQs: 50

Q1. Which network type typically covers only a few metres around an individual device?
(A) LAN (B) MAN (C) WAN (D) PAN
Answer: (D) PAN [CO-1]

Q2. Which network type is most appropriate for connecting computers within a single office building?
(A) WAN (B) LAN (C) MAN (D) PAN
Answer: (B) LAN [CO-1]

Q3. A network that spans an entire city, connecting multiple LANs, is called a:
(A) PAN (B) LAN (C) MAN (D) WAN
Answer: (C) MAN [CO-1]

Q4. The Internet is the most common example of which type of network?
(A) LAN (B) PAN (C) MAN (D) WAN
Answer: (D) WAN [CO-1]

Q5. In which topology do all devices connect to a single central cable?
(A) Star (B) Bus (C) Ring (D) Mesh
Answer: (B) Bus [CO-1]

Q6. In which topology does a failure of the central device bring down the entire network?
(A) Bus (B) Ring (C) Star (D) Mesh
Answer: (C) Star [CO-1]

Q7. Which topology requires n(n-1)/2 links for n nodes?
(A) Star (B) Bus (C) Ring (D) Mesh
Answer: (D) Mesh [CO-1]

Q8. In which topology does data travel in one direction around a closed loop?
(A) Bus (B) Star (C) Ring (D) Mesh
Answer: (C) Ring [CO-1]

Q9. A topology combining two or more different topologies is called:
(A) Mesh (B) Hybrid (C) Star (D) Bus
Answer: (B) Hybrid [CO-1]

Q10. Which topology offers the highest fault tolerance due to multiple redundant paths?
(A) Bus (B) Star (C) Ring (D) Mesh
Answer: (D) Mesh [CO-1]

Q11. In circuit switching, the communication path is:
(A) Established dynamically per packet (B) Reserved for the entire session (C) Never reserved (D) Shared randomly
Answer: (B) Reserved for the entire session [CO-1]

Q12. Which switching technique is used in the Internet (IP networks)?
(A) Circuit switching (B) Packet switching (C) Message switching (D) None of these
Answer: (B) Packet switching [CO-1]

Q13. Which of the following is a disadvantage of Circuit Switching?
(A) High reliability (B) Wastage of bandwidth during idle time (C) Low setup time (D) Dynamic routing
Answer: (B) Wastage of bandwidth during idle time [CO-1]

Q14. How many layers does the OSI reference model have?

(A) 4 (B) 5 (C) 7 (D) 6
Answer: (C) 7 [CO-1]

Q15. Which OSI layer is responsible for framing and MAC addressing?

(A) Physical (B) Data Link (C) Network (D) Transport
Answer: (B) Data Link [CO-1]

Q16. Which OSI layer is responsible for logical addressing and routing?

(A) Data Link (B) Network (C) Transport (D) Session
Answer: (B) Network [CO-1]

Q17. Which OSI layer handles end-to-end reliable delivery and segmentation?

(A) Network (B) Transport (C) Session (D) Presentation
Answer: (B) Transport [CO-1]

Q18. Which OSI layer is responsible for encryption and data compression?

(A) Session (B) Presentation (C) Application (D) Transport
Answer: (B) Presentation [CO-1]

Q19. Which OSI layer directly interacts with end-user applications like a browser?

(A) Session (B) Presentation (C) Application (D) Transport
Answer: (C) Application [CO-1]

Q20. How many layers does the TCP/IP model have?

(A) 7 (B) 5 (C) 4 (D) 6
Answer: (C) 4 [CO-1]

Q21. Which layer is present in the OSI model but combined into the Application layer in the TCP/IP model?

(A) Physical (B) Network (C) Session (D) Transport
Answer: (C) Session [CO-1]

Q22. A network switch primarily operates at which OSI layer?

(A) Physical (B) Data Link (C) Network (D) Transport
Answer: (B) Data Link [CO-1]

Q23. A router primarily operates at which OSI layer?

(A) Physical (B) Data Link (C) Network (D) Transport
Answer: (C) Network [CO-1]

Q24. Bandwidth is measured in units of:

(A) Seconds (B) Bits per second (C) Metres (D) Volts
Answer: (B) Bits per second [CO-1]

Q25. Latency is best defined as:

(A) The maximum data rate of a channel (B) The total time for a signal to travel from sender to receiver (C) The physical length of a cable (D) The number of errors in transmission
Answer: (B) The total time for a signal to travel from sender to receiver [CO-1]

Q26. Which of the following is NOT a component of end-to-end delay?

(A) Propagation delay (B) Transmission delay (C) Queuing delay (D) Signal amplitude
Answer: (D) Signal amplitude [CO-1]

Q27. The bandwidth-delay product represents:

(A) The cost of a network link (B) The amount of data that can be "in transit" on the link (C) The number of nodes in a network (D) The error rate of a channel
Answer: (B) The amount of data that can be "in transit" on the link [CO-1]

Q28. Which guided medium offers the highest bandwidth and lowest attenuation over long distances?

(A) Twisted pair (B) Coaxial cable (C) Fiber optic cable (D) None of these
Answer: (C) Fiber optic cable [CO-1]

Q29. Which guided medium is most susceptible to electromagnetic interference?

(A) Fiber optic cable (B) Twisted pair cable (C) Coaxial cable (D) None of these
Answer: (B) Twisted pair cable [CO-1]

Q30. STP cable differs from UTP cable mainly by having:

(A) No copper wires (B) An additional shielding layer (C) Higher cost only in theory (D) No twisting
Answer: (B) An additional shielding layer [CO-1]

Q31. Which of the following is an unguided transmission medium?

(A) Twisted pair (B) Coaxial cable (C) Microwave (D) Fiber optic
Answer: (C) Microwave [CO-1]

Q32. Microwave transmission requires:

(A) No line of sight (B) Line-of-sight alignment between antennas (C) Underground cabling (D) A closed loop path
Answer: (B) Line-of-sight alignment between antennas [CO-1]

Q33. Infrared communication cannot pass through:

(A) Air (B) Vacuum (C) Walls/solid obstacles (D) Glass windows only
Answer: (C) Walls/solid obstacles [CO-1]

Q34. Which unguided medium is used in TV remote controls?

(A) Radio waves (B) Microwaves (C) Infrared (D) Satellite links
Answer: (C) Infrared [CO-1]

Q35. Radio waves are generally:

(A) Directional and require line of sight (B) Omnidirectional and can penetrate walls (C) Usable only underwater (D) Limited to a few centimetres range
Answer: (B) Omnidirectional and can penetrate walls [CO-1]

Q36. Which of the following best describes a protocol in networking?

(A) A physical cable (B) A set of rules governing communication (C) A type of topology (D) A network device
Answer: (B) A set of rules governing communication [CO-1]

Q37. A repeater operates primarily to:

(A) Filter traffic based on MAC address (B) Regenerate and amplify signals (C) Translate IP addresses (D) Route packets between networks
Answer: (B) Regenerate and amplify signals [CO-1]

Q38. Which device is used to connect two different networks and forward packets based on IP address?

(A) Hub (B) Switch (C) Router (D) Repeater
Answer: (C) Router [CO-1]

Q39. A hub operates at which OSI layer?

(A) Physical (B) Data Link (C) Network (D) Transport
Answer: (A) Physical [CO-1]

Q40. Which network topology is most economical in terms of cabling for a small office?

(A) Mesh (B) Bus (C) Full Mesh (D) Ring with redundant links
Answer: (B) Bus [CO-1]

Q41. Which of the following provides the best fault isolation when a single device fails?

(A) Bus topology (B) Ring topology (C) Star topology (D) All perform equally
Answer: (C) Star topology [CO-1]

Q42. For n=6 nodes in a mesh topology, the number of links required is:

(A) 6 (B) 12 (C) 15 (D) 30
Answer: (C) 15 [CO-1]

Q43. The primary advantage of packet switching over circuit switching is:

(A) Guaranteed fixed bandwidth (B) Efficient use of shared bandwidth (C) Zero latency (D) No need for addressing
Answer: (B) Efficient use of shared bandwidth [CO-1]

Q44. In the TCP/IP model, which layer is equivalent to the OSI Network layer?

(A) Network Access layer (B) Internet layer (C) Transport layer (D) Application layer
Answer: (B) Internet layer [CO-1]

Q45. Which of the following pairs correctly matches an OSI layer with an example protocol?

(A) Application - IP (B) Transport - HTTP (C) Network - IP (D) Data Link - TCP
Answer: (C) Network - IP [CO-1]

Q46. Coaxial cable is commonly used for:

(A) Cable television distribution (B) Satellite uplink only (C) Wireless communication (D) DNS resolution
Answer: (A) Cable television distribution [CO-1]

Q47. The maximum theoretical efficiency of a network channel using guided media compared to unguided is generally:

(A) Lower due to interference (B) Higher due to lower interference and attenuation (C) The same in all cases (D) Undefined
Answer: (B) Higher due to lower interference and attenuation [CO-1]

Q48. A network administrator wants to connect 4 branch offices with the fewest cable links while retaining a single point of central management. The best topology is:

(A) Mesh (B) Star (C) Ring (D) Bus
Answer: (B) Star [CO-1]

Q49. Which of the following is true regarding fiber optic cable?

(A) It is immune to electromagnetic interference (B) It has the lowest bandwidth among guided media (C) It is cheaper than twisted pair (D) It cannot be used for long-distance links
Answer: (A) It is immune to electromagnetic interference [CO-1]

Q50. Which layer of the OSI model ensures that data from the sender's application reaches the correct application at the receiver, using port numbers?

(A) Network layer (B) Data Link layer (C) Transport layer (D) Session layer
Answer: (C) Transport layer [CO-1]

Unit-2: Addressing

Course Outcome Mapped: CO-2 | Total MCQs: 50

Q1. IPv4 address length in bits is:

(A) 16 (B) 32 (C) 64 (D) 128

Answer: (B) 32 [CO-2]

Q2. The total IPv4 address space size is:

(A) 2^16 (B) 2^24 (C) 2^32 (D) 2^64

Answer: (C) 2^32 [CO-2]

Q3. Class A default subnet mask is:

(A) 255.0.0.0 (B) 255.255.0.0 (C) 255.255.255.0 (D) 255.255.255.255

Answer: (A) 255.0.0.0 [CO-2]

Q4. Class B address range starts at:

(A) 0.0.0.0 (B) 128.0.0.0 (C) 192.0.0.0 (D) 224.0.0.0

Answer: (B) 128.0.0.0 [CO-2]

Q5. Class C default subnet mask is:

(A) 255.0.0.0 (B) 255.255.0.0 (C) 255.255.255.0 (D) 255.255.255.255

Answer: (C) 255.255.255.0 [CO-2]

Q6. Class D addresses are reserved for:

(A) Unicast (B) Multicast (C) Broadcast only (D) Private use

Answer: (B) Multicast [CO-2]

Q7. Class E addresses are reserved for:

(A) Multicast (B) Broadcast (C) Experimental/research use (D) Loopback

Answer: (C) Experimental/research use [CO-2]

Q8. Which part of an IP address is identified using the subnet mask bits set to 1?

(A) Host portion (B) Network portion (C) Broadcast portion (D) None

Answer: (B) Network portion [CO-2]

Q9. CIDR notation /26 corresponds to which subnet mask?

(A) 255.255.255.0 (B) 255.255.255.192 (C) 255.255.255.224 (D) 255.255.255.128

Answer: (B) 255.255.255.192 [CO-2]

Q10. The number of usable hosts in a /28 subnet is:

(A) 16 (B) 14 (C) 30 (D) 62

Answer: (B) 14 [CO-2]

Q11. The number of usable hosts in a /27 subnet is:

(A) 32 (B) 30 (C) 62 (D) 14

Answer: (B) 30 [CO-2]

Q12. CIDR was introduced primarily to:

(A) Increase address length (B) Eliminate class boundaries and reduce address wastage (C) Replace subnetting (D) Remove need for routers

Answer: (B) Eliminate class boundaries and reduce address wastage [CO-2]

Q13. VLSM stands for:

(A) Variable Length Subnet Mask (B) Virtual LAN Subnet Mask (C) Variable Link State Mask (D) Very Large Subnet Mask

Answer: (A) Variable Length Subnet Mask [CO-2]

Q14. The main advantage of VLSM over FLSM is:

(A) Simpler configuration (B) Efficient use of address space by allowing different subnet sizes (C) Faster routing (D) No need for subnetting

Answer: (B) Efficient use of address space by allowing different subnet sizes [CO-2]

Q15. Supernetting is used to:

(A) Split one network into many smaller ones (B) Combine multiple networks into a single larger network for route summarization (C) Encrypt IP addresses (D) Assign private IPs
Answer: (B) Combine multiple networks into a single larger network for route summarization [CO-2]

Q16. Two Class C networks 192.168.0.0/24 and 192.168.1.0/24 can be summarized into:

(A) 192.168.0.0/22 (B) 192.168.0.0/23 (C) 192.168.0.0/25 (D) 192.168.0.0/16
Answer: (B) 192.168.0.0/23 [CO-2]

Q17. NAT is primarily used to:

(A) Encrypt data (B) Translate private IP addresses to public IP addresses (C) Assign MAC addresses (D) Perform routing between ASes
Answer: (B) Translate private IP addresses to public IP addresses [CO-2]

Q18. Which NAT type maps one private address to one fixed public address permanently?

(A) Dynamic NAT (B) PAT (C) Static NAT (D) Overload NAT
Answer: (C) Static NAT [CO-2]

Q19. PAT is also known as:

(A) Static NAT (B) NAT overload (C) Dynamic NAT (D) Port forwarding only
Answer: (B) NAT overload [CO-2]

Q20. Which private IP address range belongs to Class A private addresses?

(A) 10.0.0.0/8 (B) 172.16.0.0/12 (C) 192.168.0.0/16 (D) 169.254.0.0/16
Answer: (A) 10.0.0.0/8 [CO-2]

Q21. 172.16.0.0/12 is an example of:

(A) A public IP range (B) A private IP range (C) A multicast range (D) A loopback range
Answer: (B) A private IP range [CO-2]

Q22. The loopback IP address is:

(A) 0.0.0.0 (B) 127.0.0.1 (C) 255.255.255.255 (D) 169.254.0.1
Answer: (B) 127.0.0.1 [CO-2]

Q23. Which of the following commands is used to enable NAT overload (PAT) on a Cisco router?

(A) ip nat inside source list 1 interface Gi0/1 overload (B) ip nat outside (C) ip route 0.0.0.0 (D) show running-config
Answer: (A) ip nat inside source list 1 interface Gi0/1 overload [CO-2]

Q24. Static Routing requires:

(A) Automatic route discovery (B) Manual configuration of routes by the administrator (C) No configuration (D) Dynamic protocol exchange
Answer: (B) Manual configuration of routes by the administrator [CO-2]

Q25. A Default Route is typically represented as:

(A) 255.255.255.255/32 (B) 0.0.0.0/0 (C) 127.0.0.1/8 (D) 10.0.0.0/8
Answer: (B) 0.0.0.0/0 [CO-2]

Q26. Default routing is most suitable for:

(A) Large core networks with many exit points (B) Stub networks with a single exit point (C) Data centers only (D) Multicast networks
Answer: (B) Stub networks with a single exit point [CO-2]

Q27. The broadcast address of the network 192.168.1.0/24 is:

(A) 192.168.1.1 (B) 192.168.1.254 (C) 192.168.1.255 (D) 192.168.0.255
Answer: (C) 192.168.1.255 [CO-2]

Q28. Given IP 10.10.10.65/26, the network address is:

(A) 10.10.10.0 (B) 10.10.10.64 (C) 10.10.10.128 (D) 10.10.10.63
Answer: (B) 10.10.10.64 [CO-2]

Q29. For a subnet mask of 255.255.255.240, the CIDR prefix is:

(A) /26 (B) /27 (C) /28 (D) /29
Answer: (C) /28 [CO-2]

Q30. In classful addressing, which class is intended for very large organizations with millions of hosts?

(A) Class A (B) Class B (C) Class C (D) Class D
Answer: (A) Class A [CO-2]

Q31. A network administrator needs 500 host addresses. Which subnet mask should be used?

(A) /24 (B) /23 (C) /25 (D) /22
Answer: (B) /23 [CO-2]

Q32. Which of the following is NOT a benefit of subnetting?

(A) Reduced broadcast domain size (B) Improved security (C) Increased number of usable IP addresses beyond the original block (D) Better network organization
Answer: (C) Increased number of usable IP addresses beyond the original block [CO-2]

Q33. In VLSM, subnets are typically allocated:

(A) Randomly (B) From largest host requirement to smallest (C) From smallest to largest only (D) Alphabetically
Answer: (B) From largest host requirement to smallest [CO-2]

Q34. The term 'longest prefix match' refers to:

(A) Choosing the shortest matching route (B) Choosing the most specific (longest) matching route in the routing table (C) Ignoring the subnet mask (D) Random route selection
Answer: (B) Choosing the most specific (longest) matching route in the routing table [CO-2]

Q35. Dynamic NAT differs from Static NAT because it:

(A) Maps addresses one-to-one permanently (B) Maps private addresses to any available address from a pool (C) Does not require any configuration (D) Only works with IPv6
Answer: (B) Maps private addresses to any available address from a pool [CO-2]

Q36. Which of the following is a valid classless IP address representation?

(A) 192.168.1.0 Class C (B) 192.168.1.0/26 (C) Class A 10.0.0.0 (D) None of these
Answer: (B) 192.168.1.0/26 [CO-2]

Q37. The subnet mask determines:

(A) The MAC address of a host (B) Which portion of an IP address is the network ID and which is the host ID (C) The physical topology (D) The transmission medium
Answer: (B) Which portion of an IP address is the network ID and which is the host ID [CO-2]

Q38. What is the maximum number of usable hosts in a Class C network without subnetting?

(A) 256 (B) 254 (C) 255 (D) 64
Answer: (B) 254 [CO-2]

Q39. The 'access-list' command in NAT configuration is used to:

(A) Define routing protocols (B) Specify which internal addresses are eligible for translation (C) Assign VLANs (D) Configure OSPF areas
Answer: (B) Specify which internal addresses are eligible for translation [CO-2]

Q40. For the network 192.168.10.0/24 divided using VLSM for a subnet needing 50 hosts, the most efficient subnet mask is:

(A) /24 (B) /25 (C) /26 (D) /27
Answer: (C) /26 [CO-2]

Q41. Which command displays active NAT translations on a Cisco router?

(A) show ip route (B) show ip nat translations (C) show running-config (D) show interfaces
Answer: (B) show ip nat translations [CO-2]

Q42. IPv4 addresses are exhausted primarily due to:

(A) Too few classes (B) Inefficient classful allocation and explosive Internet growth (C) Use of VLSM (D) Overuse of loopback addresses

Answer: (B) Inefficient classful allocation and explosive Internet growth [CO-2]

Q43. A /30 subnet is commonly used for:

(A) LAN with 100 hosts (B) Point-to-point WAN links between two routers (C) Broadcast domains (D) Multicast groups

Answer: (B) Point-to-point WAN links between two routers [CO-2]

Q44. What does the 'inside source' keyword specify in a NAT configuration?

(A) Outside public address translation (B) Which inside addresses are translated when going outside (C) VLAN tagging (D) Routing protocol

Answer: (B) Which inside addresses are translated when going outside [CO-2]

Q45. The network 200.1.1.0/24 needs to be divided into 4 equal subnets. What is the new subnet mask?

(A) /25 (B) /26 (C) /27 (D) /28

Answer: (B) /26 [CO-2]

Q46. Which of these addresses is a valid host address on the network 192.168.1.0/28?

(A) 192.168.1.0 (B) 192.168.1.15 (C) 192.168.1.5 (D) 192.168.1.16

Answer: (C) 192.168.1.5 [CO-2]

Q47. NAT is generally considered a:

(A) Layer 2 function (B) Layer 3 function performed typically at the network edge (C) Application layer protocol (D) Physical layer function

Answer: (B) Layer 3 function performed typically at the network edge [CO-2]

Q48. Which of the following best describes classful addressing's main drawback?

(A) Too complex to configure (B) Inflexible fixed boundaries leading to wasted address space (C) Cannot support routing (D) Requires IPv6

Answer: (B) Inflexible fixed boundaries leading to wasted address space [CO-2]

Q49. In VLSM design, if a subnet needs only 2 usable host addresses (e.g., point-to-point link), which mask is appropriate?

(A) /24 (B) /28 (C) /30 (D) /16

Answer: (C) /30 [CO-2]

Q50. Which of the following is true about a supernet?

(A) It always uses a longer prefix than its component networks (B) It always uses a shorter prefix than its component networks (C) It requires NAT (D) It only works with IPv6

Answer: (B) It always uses a shorter prefix than its component networks [CO-2]

Unit-3: Routing

Course Outcome Mapped: CO-3 | Total MCQs: 50

Q1. Which routing algorithm is based on the Bellman-Ford algorithm?

(A) Link State (B) Distance Vector (C) Path Vector (D) None

Answer: (B) Distance Vector [CO-3]

Q2. Which routing algorithm requires each router to build a complete map of the network topology?

(A) Distance Vector (B) Link State (C) Path Vector (D) Static

Answer: (B) Link State [CO-3]

Q3. RIP is an example of which routing algorithm type?

(A) Link State (B) Distance Vector (C) Path Vector (D) Hybrid only

Answer: (B) Distance Vector [CO-3]

Q4. OSPF is an example of which routing algorithm type?

(A) Distance Vector (B) Link State (C) Path Vector (D) None

Answer: (B) Link State [CO-3]

Q5. BGP is an example of which routing algorithm type?

(A) Distance Vector (B) Link State (C) Path Vector (D) Static

Answer: (C) Path Vector [CO-3]

Q6. The maximum hop count supported by RIP is:

(A) 15 (B) 16 (C) 255 (D) 30

Answer: (A) 15 [CO-3]

Q7. RIP updates its routing table every:

(A) 10 seconds (B) 30 seconds (C) 60 seconds (D) 90 seconds

Answer: (B) 30 seconds [CO-3]

Q8. Which version of RIP supports VLSM/CIDR?

(A) RIP V1 (B) RIP V2 (C) Both equally (D) Neither

Answer: (B) RIP V2 [CO-3]

Q9. RIP V2 sends updates using:

(A) Broadcast (B) Multicast address 224.0.0.9 (C) Unicast only (D) Anycast

Answer: (B) Multicast address 224.0.0.9 [CO-3]

Q10. OSPF uses which algorithm to compute the shortest path?

(A) Bellman-Ford (B) Dijkstra's algorithm (C) DUAL (D) Distance vector

Answer: (B) Dijkstra's algorithm [CO-3]

Q11. In OSPF, which area must always be present?

(A) Area 1 (B) Area 0 (Backbone) (C) Area 51 (D) Stub area

Answer: (B) Area 0 (Backbone) [CO-3]

Q12. What does LSA stand for in OSPF?

(A) Link State Algorithm (B) Link State Advertisement (C) Local Subnet Address (D) Least Significant Address

Answer: (B) Link State Advertisement [CO-3]

Q13. EIGRP uses which algorithm for loop-free fast convergence?

(A) Dijkstra (B) Bellman-Ford (C) DUAL (D) RSA

Answer: (C) DUAL [CO-3]

Q14. EIGRP is:

(A) An open industry standard (B) A Cisco-proprietary protocol (C) A path vector protocol (D) Used only for exterior routing

Answer: (B) A Cisco-proprietary protocol [CO-3]

Q15. BGP is primarily used for:

(A) Routing within a single organization (B) Routing between different autonomous systems on the Internet (C) Routing within a LAN (D) Wireless access

Answer: (B) Routing between different autonomous systems on the Internet [CO-3]

Q16. The AS-PATH attribute in BGP is used to:

(A) Encrypt routes (B) Prevent routing loops between autonomous systems (C) Compress routing tables (D) Assign IP addresses

Answer: (B) Prevent routing loops between autonomous systems [CO-3]

Q17. IPv6 addresses are how many bits long?

(A) 32 (B) 64 (C) 128 (D) 256

Answer: (C) 128 [CO-3]

Q18. IPv6 addresses are typically written in:

(A) Dotted decimal (B) Eight groups of hexadecimal digits separated by colons (C) Binary only (D) Four octets

Answer: (B) Eight groups of hexadecimal digits separated by colons [CO-3]

Q19. The '::' notation in an IPv6 address is used to:

(A) Indicate a broadcast address (B) Compress one or more groups of consecutive zeros (C) Separate network and host (D) Denote a multicast group

Answer: (B) Compress one or more groups of consecutive zeros [CO-3]

Q20. Which of the following is NOT a type of IPv6 address?

(A) Unicast (B) Multicast (C) Anycast (D) Broadcast

Answer: (D) Broadcast [CO-3]

Q21. Multicasting in IPv4 uses which class of addresses?

(A) Class A (B) Class B (C) Class C (D) Class D

Answer: (D) Class D [CO-3]

Q22. Which protocol is used by hosts to join a multicast group in IPv4?

(A) ARP (B) IGMP (C) ICMP (D) DHCP

Answer: (B) IGMP [CO-3]

Q23. Default Routing is best suited for:

(A) Core routers with multiple paths (B) Stub networks with a single exit point (C) Multicast domains (D) Data centers with load balancing

Answer: (B) Stub networks with a single exit point [CO-3]

Q24. Which routing type requires no manual configuration and adapts automatically to topology changes?

(A) Static routing (B) Dynamic routing (C) Default routing (D) None

Answer: (B) Dynamic routing [CO-3]

Q25. What is the term for the time taken by all routers to update their tables after a topology change?

(A) Latency (B) Convergence time (C) Propagation delay (D) Throughput

Answer: (B) Convergence time [CO-3]

Q26. Which of the following is a disadvantage of Distance Vector routing?

(A) Fast convergence (B) Slow convergence and susceptibility to routing loops (C) Low CPU usage (D) High scalability

Answer: (B) Slow convergence and susceptibility to routing loops [CO-3]

Q27. Split horizon is a technique used to:

(A) Speed up convergence (B) Prevent a router from advertising a route back to the neighbour it learned it from (C) Encrypt routing updates (D) Reduce hop count

Answer: (B) Prevent a router from advertising a route back to the neighbour it learned it from [CO-3]

Q28. Which routing protocol metric is purely hop-count based?

(A) OSPF (B) RIP (C) EIGRP (D) BGP

Answer: (B) RIP [CO-3]

Q29. OSPF calculates its cost metric primarily based on:

(A) Hop count (B) Interface bandwidth (C) Number of ASes (D) Delay only
Answer: (B) Interface bandwidth [CO-3]

Q30. Which protocol exchanges 'Hello' packets to discover neighbours?

(A) RIP (B) OSPF (C) Static routing (D) NAT
Answer: (B) OSPF [CO-3]

Q31. What is an Autonomous System (AS) in the context of BGP?

(A) A single router (B) A collection of networks under a single administrative control (C) A type of IP address (D) A subnet
Answer: (B) A collection of networks under a single administrative control [CO-3]

Q32. eBGP is used to exchange routes between:

(A) Routers within the same AS (B) Routers in different ASes (C) Only static routers (D) Multicast groups
Answer: (B) Routers in different ASes [CO-3]

Q33. iBGP is used to exchange routes:

(A) Between different ASes (B) Within the same AS (C) Only over satellite links (D) Only for IPv6
Answer: (B) Within the same AS [CO-3]

Q34. Which of the following best describes 'administrative distance'?

(A) A metric used to measure bandwidth (B) A value used to rate the trustworthiness of a routing source (C) The physical distance between routers (D) The number of hops
Answer: (B) A value used to rate the trustworthiness of a routing source [CO-3]

Q35. Which routing protocol generally has the lowest (most trusted) administrative distance among dynamic protocols?

(A) RIP (B) OSPF (C) EIGRP (D) BGP
Answer: (C) EIGRP [CO-3]

Q36. Route summarization primarily helps to:

(A) Increase routing table size (B) Reduce routing table size and improve scalability (C) Slow down convergence (D) Disable dynamic routing
Answer: (B) Reduce routing table size and improve scalability [CO-3]

Q37. Which of these is an Interior Gateway Protocol (IGP)?

(A) BGP (B) OSPF (C) Both (D) Neither
Answer: (B) OSPF [CO-3]

Q38. Which of these is an Exterior Gateway Protocol (EGP)?

(A) RIP (B) OSPF (C) BGP (D) EIGRP
Answer: (C) BGP [CO-3]

Q39. In Link State Routing, routers exchange information about:

(A) Entire routing tables (B) Only their directly connected links (C) Only their IP addresses (D) Nothing; they compute paths independently
Answer: (B) Only their directly connected links [CO-3]

Q40. Path Vector routing avoids loops by:

(A) Using hop count limits (B) Checking whether the router's own AS already appears in the advertised path (C) Using split horizon only (D) Ignoring loop detection
Answer: (B) Checking whether the router's own AS already appears in the advertised path [CO-3]

Q41. Which protocol uses 'areas' to improve scalability in large networks?

(A) RIP (B) OSPF (C) BGP (D) Static routing
Answer: (B) OSPF [CO-3]

Q42. The purpose of a routing table's 'next hop' field is to:

(A) Indicate the destination network (B) Indicate the next router to forward the packet to (C) Indicate the interface bandwidth (D) Indicate the AS number

Answer: (B) Indicate the next router to forward the packet to [CO-3]

Q43. Which routing protocol type is best suited for very large, Internet-scale routing between organizations?

(A) Distance Vector (B) Link State (C) Path Vector (D) None

Answer: (C) Path Vector [CO-3]

Q44. What triggers a routing update in Link State Routing (compared to periodic updates in Distance Vector)?

(A) Fixed timer only (B) A change in the network topology (event-driven) (C) Random intervals (D) Never updated

Answer: (B) A change in the network topology (event-driven) [CO-3]

Q45. IPv6 was primarily developed to solve which major problem of IPv4?

(A) Slow routing (B) Address exhaustion (C) Lack of subnetting (D) High latency

Answer: (B) Address exhaustion [CO-3]

Q46. A multicast address in IPv4 identifies:

(A) A single specific host (B) A group of interested receivers (C) The nearest host in a group (D) A broadcast domain

Answer: (B) A group of interested receivers [CO-3]

Q47. Which of the following correctly orders routing protocols from oldest/simplest to most policy-driven?

(A) BGP, OSPF, RIP (B) RIP, OSPF, BGP (C) OSPF, RIP, BGP (D) BGP, RIP, OSPF

Answer: (B) RIP, OSPF, BGP [CO-3]

Q48. Which statement about EIGRP's metric is correct?

(A) Uses only hop count (B) Uses a composite metric based on bandwidth, delay, reliability and load (C) Uses only delay (D) Uses only AS-PATH

Answer: (B) Uses a composite metric based on bandwidth, delay, reliability and load [CO-3]

Q49. In Distance Vector routing, a router updates its table when:

(A) It receives a shorter path advertisement from a neighbour (B) It restarts (C) It loses power (D) Never updates

Answer: (A) It receives a shorter path advertisement from a neighbour [CO-3]

Q50. Which of the following is true about static routing in a network with frequently changing topology?

(A) It adapts automatically (B) It requires manual reconfiguration for every change, making it unsuitable (C) It is always preferred (D) It uses Dijkstra's algorithm

Answer: (B) It requires manual reconfiguration for every change, making it unsuitable [CO-3]

Unit-4: Medium Access Control

Course Outcome Mapped: CO-4 | Total MCQs: 50

Q1. The maximum efficiency of Pure ALOHA is approximately:

(A) 18.4% (B) 36.8% (C) 50% (D) 100%

Answer: (A) 18.4% [CO-4]

Q2. The maximum efficiency of Slotted ALOHA is approximately:

(A) 18.4% (B) 36.8% (C) 50% (D) 90%

Answer: (B) 36.8% [CO-4]

Q3. In Slotted ALOHA, transmissions can only begin:

(A) At any time (B) At the start of a fixed time slot (C) Only after collision (D) Never

Answer: (B) At the start of a fixed time slot [CO-4]

Q4. CSMA stands for:

(A) Carrier Sense Multiple Access (B) Collision Sense Multiple Access (C) Channel Sense Media Access (D) Carrier Signal Multiple Access

Answer: (A) Carrier Sense Multiple Access [CO-4]

Q5. CSMA/CD is primarily used in:

(A) Wireless LANs (B) Traditional wired Ethernet (C) Satellite links (D) Fiber optic backbones only

Answer: (B) Traditional wired Ethernet [CO-4]

Q6. CSMA/CA is primarily used in:

(A) Wired Ethernet (B) Wireless LANs (IEEE 802.11) (C) Token Ring (D) ATM networks

Answer: (B) Wireless LANs (IEEE 802.11) [CO-4]

Q7. What does a station do immediately after detecting a collision in CSMA/CD?

(A) Continues transmitting (B) Sends a jam signal and stops transmitting (C) Ignores it (D) Switches to CSMA/CA

Answer: (B) Sends a jam signal and stops transmitting [CO-4]

Q8. The backoff algorithm used in CSMA/CD is called:

(A) Linear backoff (B) Binary exponential backoff (C) Random walk (D) Fixed backoff

Answer: (B) Binary exponential backoff [CO-4]

Q9. The hidden terminal problem occurs mainly in:

(A) Wired networks (B) Wireless networks (C) Fiber optic networks (D) Point-to-point links

Answer: (B) Wireless networks [CO-4]

Q10. RTS/CTS frames in CSMA/CA are used to:

(A) Detect collisions after they occur (B) Reserve the channel before data transmission (C) Encrypt data (D) Perform error correction

Answer: (B) Reserve the channel before data transmission [CO-4]

Q11. IEEE 802.3 standard refers to:

(A) Token Ring (B) Ethernet (C) Wireless LAN (D) Bluetooth

Answer: (B) Ethernet [CO-4]

Q12. IEEE 802.5 standard refers to:

(A) Ethernet (B) Token Ring (C) Wi-Fi (D) PPP

Answer: (B) Token Ring [CO-4]

Q13. In Token Ring, a station may transmit only when it:

(A) Senses the channel is idle (B) Possesses the token (C) Detects a collision (D) Receives a jam signal

Answer: (B) Possesses the token [CO-4]

Q14. Compared to Ethernet, Token Ring provides:

(A) Random access with possible collisions (B) Deterministic, collision-free access (C) No access control (D) Only broadcast transmission

Answer: (B) Deterministic, collision-free access [CO-4]

Q15. In Stop-and-Wait flow control, the sender:

(A) Sends multiple frames before waiting (B) Sends one frame and waits for an ACK before sending the next (C) Never waits for ACK (D) Sends frames randomly

Answer: (B) Sends one frame and waits for an ACK before sending the next [CO-4]

Q16. The main drawback of Stop-and-Wait flow control is:

(A) High complexity (B) Poor channel utilization on high bandwidth-delay links (C) No error detection (D) Too many collisions

Answer: (B) Poor channel utilization on high bandwidth-delay links [CO-4]

Q17. Sliding Window flow control improves efficiency by:

(A) Sending only one frame at a time (B) Allowing multiple frames to be sent before requiring an acknowledgement (C) Ignoring acknowledgements (D) Using larger frames only

Answer: (B) Allowing multiple frames to be sent before requiring an acknowledgement [CO-4]

Q18. In Stop-and-Wait ARQ, retransmission occurs when:

(A) The receiver requests it explicitly only (B) A timer expires without receiving an ACK (C) The sender decides randomly (D) Never

Answer: (B) A timer expires without receiving an ACK [CO-4]

Q19. Go-Back-N ARQ, upon detecting an error, retransmits:

(A) Only the erroneous frame (B) The erroneous frame and all subsequent frames sent (C) No frames (D) All frames from the beginning of the session

Answer: (B) The erroneous frame and all subsequent frames sent [CO-4]

Q20. Selective Repeat ARQ, upon detecting an error, retransmits:

(A) All frames (B) Only the specific erroneous frame(s) (C) The entire window (D) Nothing

Answer: (B) Only the specific erroneous frame(s) [CO-4]

Q21. Which ARQ protocol requires more buffering at the receiver?

(A) Stop-and-Wait ARQ (B) Go-Back-N ARQ (C) Selective Repeat ARQ (D) None require buffering

Answer: (C) Selective Repeat ARQ [CO-4]

Q22. A single parity bit can reliably detect:

(A) Any number of bit errors (B) Only single-bit (odd number of) errors (C) Only even-bit errors (D) No errors

Answer: (B) Only single-bit (odd number of) errors [CO-4]

Q23. In checksum error detection, the receiver considers data error-free if the sum (including checksum) results in:

(A) All zeros (B) All ones (C) A prime number (D) Exactly the original checksum

Answer: (B) All ones [CO-4]

Q24. CRC (Cyclic Redundancy Check) treats data as:

(A) A fixed matrix (B) A polynomial divided by a generator polynomial (C) An encrypted string (D) A hash table

Answer: (B) A polynomial divided by a generator polynomial [CO-4]

Q25. In CRC, an error is detected at the receiver if:

(A) The remainder is zero (B) The remainder is non-zero (C) The quotient is zero (D) The generator polynomial changes

Answer: (B) The remainder is non-zero [CO-4]

Q26. Hamming code is primarily designed to:

(A) Only detect errors (B) Detect and correct single-bit errors (C) Compress data (D) Encrypt data

Answer: (B) Detect and correct single-bit errors [CO-4]

Q27. In Hamming code, redundant bits are placed at positions that are:

(A) Multiples of 3 (B) Powers of 2 (1, 2, 4, 8...) (C) Prime numbers (D) Even numbers only
Answer: (B) Powers of 2 (1, 2, 4, 8...) [CO-4]

Q28. For 4 data bits, the minimum number of Hamming redundant bits required is:

(A) 2 (B) 3 (C) 4 (D) 5
Answer: (B) 3 [CO-4]

Q29. HDLC stands for:

(A) High-level Data Link Control (B) Host Data Link Configuration (C) High Data Loop Control (D) Hybrid Data Link Communication
Answer: (A) High-level Data Link Control [CO-4]

Q30. HDLC uses which type of framing?

(A) Byte-oriented (B) Bit-oriented with flag bytes and bit stuffing (C) Character-oriented only (D) No framing
Answer: (B) Bit-oriented with flag bytes and bit stuffing [CO-4]

Q31. PPP stands for:

(A) Point-to-Point Protocol (B) Packet Processing Protocol (C) Private Path Protocol (D) Public Peer Protocol
Answer: (A) Point-to-Point Protocol [CO-4]

Q32. Which sub-protocol of PPP is responsible for establishing and configuring the link?

(A) NCP (B) LCP (C) ARP (D) IGMP
Answer: (B) LCP [CO-4]

Q33. Which sub-protocol of PPP is responsible for configuring different network layer protocols?

(A) LCP (B) NCP (C) DUAL (D) RTS
Answer: (B) NCP [CO-4]

Q34. PPP supports authentication using:

(A) Only PAP (B) Only CHAP (C) Both PAP and CHAP (D) Neither
Answer: (C) Both PAP and CHAP [CO-4]

Q35. Compared to HDLC, PPP is:

(A) Vendor-specific and less standardized (B) More standardized and interoperable for WAN point-to-point links (C) Only used in LANs (D) Obsolete
Answer: (B) More standardized and interoperable for WAN point-to-point links [CO-4]

Q36. The 'vulnerable period' in Pure ALOHA (in terms of frame transmission time Tfr) is:

(A) Tfr (B) 2 x Tfr (C) 0.5 x Tfr (D) 4 x Tfr
Answer: (B) 2 x Tfr [CO-4]

Q37. The vulnerable period in Slotted ALOHA is:

(A) Tfr (B) 2 x Tfr (C) 0.5 x Tfr (D) 3 x Tfr
Answer: (A) Tfr [CO-4]

Q38. Which error control mechanism is generally more bandwidth-efficient on long, high-error-rate links?

(A) Stop-and-Wait ARQ (B) Selective Repeat ARQ (C) Neither (D) Both equally
Answer: (B) Selective Repeat ARQ [CO-4]

Q39. In two-dimensional parity check, parity bits are computed:

(A) Only for rows (B) Only for columns (C) For both rows and columns (D) Neither
Answer: (C) For both rows and columns [CO-4]

Q40. Which of these best distinguishes flow control from error control?

(A) They are the same thing (B) Flow control manages the rate of data transmission; error control manages detection/correction of transmission errors (C) Flow control only applies to wireless (D) Error control only applies to Ethernet
Answer: (B) Flow control manages the rate of data transmission; error control manages detection/correction of transmission errors [CO-4]

Q41. A jam signal in CSMA/CD is sent to:

(A) Encrypt the frame (B) Ensure all stations are aware a collision has occurred (C) Request retransmission from the receiver (D) Switch to a different channel

Answer: (B) Ensure all stations are aware a collision has occurred [CO-4]

Q42. Which flow control technique is equivalent to Sliding Window with a window size of 1?

(A) Go-Back-N (B) Selective Repeat (C) Stop-and-Wait (D) CSMA/CD

Answer: (C) Stop-and-Wait [CO-4]

Q43. In CSMA/CA, after sensing the channel is idle, a station:

(A) Transmits immediately (B) Waits an additional random backoff time before transmitting (C) Sends a jam signal (D) Switches to CSMA/CD

Answer: (B) Waits an additional random backoff time before transmitting [CO-4]

Q44. Bit stuffing in HDLC is used to:

(A) Increase transmission speed (B) Prevent the data from being mistaken for a flag byte (C) Perform error correction (D) Compress data

Answer: (B) Prevent the data from being mistaken for a flag byte [CO-4]

Q45. Ethernet frames are primarily identified/delimited using:

(A) Token passing (B) Preamble and Start Frame Delimiter (C) Bit stuffing only (D) Parity bits

Answer: (B) Preamble and Start Frame Delimiter [CO-4]

Q46. Which of these protocols does NOT provide error control (only flow control)?

(A) Stop-and-Wait ARQ (B) Go-Back-N ARQ (C) Basic Stop-and-Wait flow control (without ARQ) (D) Selective Repeat ARQ

Answer: (C) Basic Stop-and-Wait flow control (without ARQ) [CO-4]

Q47. CRC provides better protection than a simple checksum mainly because it:

(A) Uses fewer bits (B) Detects burst errors more reliably (C) Requires no computation (D) Works only for text data

Answer: (B) Detects burst errors more reliably [CO-4]

Q48. In a Hamming code with codeword length 7 (4 data + 3 parity), the code is referred to as:

(A) Hamming(4,3) (B) Hamming(7,4) (C) Hamming(3,4) (D) Hamming(4,7)

Answer: (B) Hamming(7,4) [CO-4]

Q49. Which access method does modern switched (full-duplex) Ethernet primarily rely on instead of CSMA/CD?

(A) Token passing (B) Dedicated point-to-point links without contention (C) ALOHA (D) CSMA/CA

Answer: (B) Dedicated point-to-point links without contention [CO-4]

Q50. Which of the following data link protocols is most suitable for a simple point-to-point leased line connecting two routers with authentication support?

(A) HDLC (B) PPP (C) Token Ring (D) CSMA/CD

Answer: (B) PPP [CO-4]

Unit-5: Transport and Application Layer Protocols

Course Outcome Mapped: CO-5 | Total MCQs: 50

Q1. Port numbers are how many bits long?

(A) 8 (B) 16 (C) 32 (D) 64

Answer: (B) 16 [CO-5]

Q2. The range of well-known ports is:

(A) 0-1023 (B) 1024-49151 (C) 49152-65535 (D) 0-65535

Answer: (A) 0-1023 [CO-5]

Q3. HTTP uses which default port number?

(A) 21 (B) 23 (C) 80 (D) 443

Answer: (C) 80 [CO-5]

Q4. HTTPS uses which default port number?

(A) 80 (B) 443 (C) 25 (D) 53

Answer: (B) 443 [CO-5]

Q5. FTP control connection uses which default port?

(A) 20 (B) 21 (C) 23 (D) 25

Answer: (B) 21 [CO-5]

Q6. SMTP uses which default port number?

(A) 21 (B) 23 (C) 25 (D) 110

Answer: (C) 25 [CO-5]

Q7. DNS uses which default port number?

(A) 21 (B) 25 (C) 53 (D) 80

Answer: (C) 53 [CO-5]

Q8. POP3 uses which default port number?

(A) 110 (B) 143 (C) 25 (D) 53

Answer: (A) 110 [CO-5]

Q9. IMAP uses which default port number?

(A) 110 (B) 143 (C) 25 (D) 53

Answer: (B) 143 [CO-5]

Q10. Telnet uses which default port number?

(A) 21 (B) 22 (C) 23 (D) 25

Answer: (C) 23 [CO-5]

Q11. A socket is formed by combining:

(A) Two IP addresses (B) An IP address and a port number (C) Two port numbers (D) A MAC address and IP address

Answer: (B) An IP address and a port number [CO-5]

Q12. Which transport protocol is connectionless?

(A) TCP (B) UDP (C) Both (D) Neither

Answer: (B) UDP [CO-5]

Q13. Which transport protocol provides reliable, ordered delivery?

(A) UDP (B) TCP (C) Both equally (D) Neither

Answer: (B) TCP [CO-5]

Q14. The minimum UDP header size is:

(A) 8 bytes (B) 20 bytes (C) 12 bytes (D) 16 bytes

Answer: (A) 8 bytes [CO-5]

Q15. The minimum TCP header size is:

(A) 8 bytes (B) 12 bytes (C) 20 bytes (D) 32 bytes
Answer: (C) 20 bytes [CO-5]

Q16. Which of these applications typically prefers UDP over TCP?

(A) File transfer (B) Email (C) DNS queries and video streaming (D) Web page loading
Answer: (C) DNS queries and video streaming [CO-5]

Q17. The TCP three-way handshake uses which sequence of flags?

(A) SYN, SYN, ACK (B) SYN, SYN-ACK, ACK (C) ACK, SYN, FIN (D) FIN, ACK, SYN
Answer: (B) SYN, SYN-ACK, ACK [CO-5]

Q18. TCP connection termination typically uses a:

(A) Two-way handshake (B) Three-way handshake (C) Four-way handshake (D) No handshake
Answer: (C) Four-way handshake [CO-5]

Q19. Which TCP header field is used for flow control?

(A) Sequence number (B) Acknowledgement number (C) Window size (D) Checksum
Answer: (C) Window size [CO-5]

Q20. Which TCP flag is used to initiate a connection?

(A) ACK (B) SYN (C) FIN (D) RST
Answer: (B) SYN [CO-5]

Q21. Which TCP flag is used to gracefully terminate a connection?

(A) SYN (B) ACK (C) FIN (D) URG
Answer: (C) FIN [CO-5]

Q22. HTTP operates on which model?

(A) Peer-to-peer (B) Request-response (client-server) (C) Broadcast (D) Multicast
Answer: (B) Request-response (client-server) [CO-5]

Q23. Which HTTP method is used to retrieve data from a server?

(A) POST (B) GET (C) DELETE (D) PUT
Answer: (B) GET [CO-5]

Q24. Which HTTP method is used to submit data to be processed to a server?

(A) GET (B) POST (C) HEAD (D) OPTIONS
Answer: (B) POST [CO-5]

Q25. HTTP status code 404 indicates:

(A) Success (B) Server error (C) Resource not found (D) Redirection
Answer: (C) Resource not found [CO-5]

Q26. HTTP status code 200 indicates:

(A) Not found (B) Success (OK) (C) Server error (D) Redirection
Answer: (B) Success (OK) [CO-5]

Q27. HTTPS differs from HTTP mainly by adding:

(A) Faster speed only (B) Encryption via SSL/TLS (C) A new port only (D) Compression
Answer: (B) Encryption via SSL/TLS [CO-5]

Q28. FTP uses how many separate TCP connections for control and data?

(A) One (B) Two (C) Three (D) Four
Answer: (B) Two [CO-5]

Q29. In FTP Active mode, the data connection is initiated by:

(A) The client (B) The server (C) Neither (D) Both simultaneously
Answer: (B) The server [CO-5]

Q30. In FTP Passive mode, the data connection is initiated by:

(A) The server (B) The client (C) Neither (D) A third party
Answer: (B) The client [CO-5]

Q31. Which protocol is used to send/relay outgoing email between mail servers?

(A) POP3 (B) IMAP (C) SMTP (D) Telnet
Answer: (C) SMTP [CO-5]

Q32. Which protocol allows a client to manage mail directly on the server, keeping it synchronized across devices?

(A) POP3 (B) IMAP (C) SMTP (D) FTP
Answer: (B) IMAP [CO-5]

Q33. Which protocol typically downloads mail to a single device and removes it from the server?

(A) IMAP (B) POP3 (C) SMTP (D) DNS
Answer: (B) POP3 [CO-5]

Q34. Telnet provides:

(A) An encrypted terminal session (B) A plain-text remote terminal session (C) File transfer only (D) DNS resolution
Answer: (B) A plain-text remote terminal session [CO-5]

Q35. Telnet has largely been replaced by which more secure protocol?

(A) FTP (B) SSH (C) HTTP (D) SMTP
Answer: (B) SSH [CO-5]

Q36. DNS translates:

(A) IP addresses to MAC addresses (B) Domain names to IP addresses (C) Port numbers to IP addresses (D) MAC addresses to domain names
Answer: (B) Domain names to IP addresses [CO-5]

Q37. The top level of the DNS hierarchy is:

(A) The TLD servers (B) The root servers (C) The authoritative servers (D) The local resolver
Answer: (B) The root servers [CO-5]

Q38. Which DNS record maps a domain name to an IPv4 address?

(A) MX (B) A (C) CNAME (D) NS
Answer: (B) A [CO-5]

Q39. Which DNS record specifies the mail server for a domain?

(A) A (B) MX (C) CNAME (D) PTR
Answer: (B) MX [CO-5]

Q40. A DNS query where the resolver does all the work of contacting each server on the client's behalf is called:

(A) Iterative query (B) Recursive query (C) Static query (D) Reverse query
Answer: (B) Recursive query [CO-5]

Q41. A DNS query where the server responds with a referral to another server instead of the final answer is called:

(A) Recursive (B) Iterative (C) Reverse (D) Cached
Answer: (B) Iterative [CO-5]

Q42. DNS caching is used to:

(A) Slow down repeated lookups (B) Improve performance by storing previous lookup results (C) Encrypt DNS queries (D) Replace the need for a resolver
Answer: (B) Improve performance by storing previous lookup results [CO-5]

Q43. Which layer is responsible for multiplexing/demultiplexing data to the correct application using port numbers?

(A) Network layer (B) Transport layer (C) Data Link layer (D) Physical layer
Answer: (B) Transport layer [CO-5]

Q44. TCP's Maximum Segment Size (MSS) refers to:

(A) The maximum size of a TCP header (B) The largest amount of data that can be sent in a single TCP segment (C) The number of retransmissions allowed (D) The window size limit

Answer: (B) The largest amount of data that can be sent in a single TCP segment [CO-5]

Q45. TCP achieves reliability over an unreliable IP network mainly through:

(A) Encryption (B) Acknowledgements and retransmissions (C) Broadcasting (D) Compression

Answer: (B) Acknowledgements and retransmissions [CO-5]

Q46. A persistent HTTP connection allows:

(A) Only one request per TCP connection (B) Multiple requests/responses over a single TCP connection (C) No TCP connection at all (D) Only UDP-based transfer

Answer: (B) Multiple requests/responses over a single TCP connection [CO-5]

Q47. HTTP cookies are primarily used to:

(A) Encrypt HTTP traffic (B) Maintain state/session information between a stateless client and server (C) Replace DNS (D) Compress web pages

Answer: (B) Maintain state/session information between a stateless client and server [CO-5]

Q48. The World Wide Web (WWW) is best described as:

(A) The same as the Internet (B) A system of interlinked hypertext documents accessed via the Internet (C) A type of transport protocol (D) A hardware device

Answer: (B) A system of interlinked hypertext documents accessed via the Internet [CO-5]

Q49. Which of the following correctly matches protocol to primary transport layer protocol used?

(A) DNS - always TCP only (B) HTTP - typically TCP (C) FTP - always UDP (D) SMTP - always UDP

Answer: (B) HTTP - typically TCP [CO-5]

Q50. The complete resolution of a URL request typically involves, in order:

(A) HTTP request, then DNS lookup, then TCP connection (B) DNS lookup, then TCP connection, then HTTP request (C) TCP connection, then HTTP request, then DNS lookup (D) DNS lookup, then HTTP request, then TCP connection

Answer: (B) DNS lookup, then TCP connection, then HTTP request [CO-5]